Lab 3: Component Modelling & Architectural Pattern Selection

16-09-2026

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Scenario: Self-Service Coffee Kiosk System

Architectural Selection: We chose the Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Style Analysis

Layered architecture separates the system into presentation, business, and data-related parts. Microservices can scale services independently, but they add network communication and operational complexity that is not necessary for a small kiosk. Client-server gives centralized control, but it can create a single point of failure and a scalability bottleneck. Layered architecture fits the kiosk because its functions are fairly simple and can be separated clearly.

Two Reasons for the Choice

1. The touch screen, ordering logic, payment processing, and stored menu information can be kept in separate layers, making the system easier to understand and maintain.

2. The kiosk has a limited set of functions, so a simpler architecture avoids the extra deployment and communication overhead of microservices.

Main Components

The five main components are: Touch Screen Interface, Order Manager, Payment Service, Menu & Pricing Database, and Receipt Printer Interface. The Order Manager handles the order and connects the other required services.

Security Advantage

The layered structure separates the user interface from payment and data operations. Payment requests can be handled through a controlled service instead of allowing the touch screen to access payment or stored data directly. This reduces unnecessary access between components.

Performance Benefit

The architecture keeps communication between the main components simple and local. Menu data can also be retrieved directly from the data layer, reducing unnecessary communication and helping the kiosk respond quickly to touch screen interactions and order processing.

Key Interfaces: Touch Screen → Order Manager (API calls); Order Manager → Payment Service (payment processing requests); Order Manager → Menu & Pricing Database (database queries); Order Manager → Receipt Printer Interface (hardware interface).